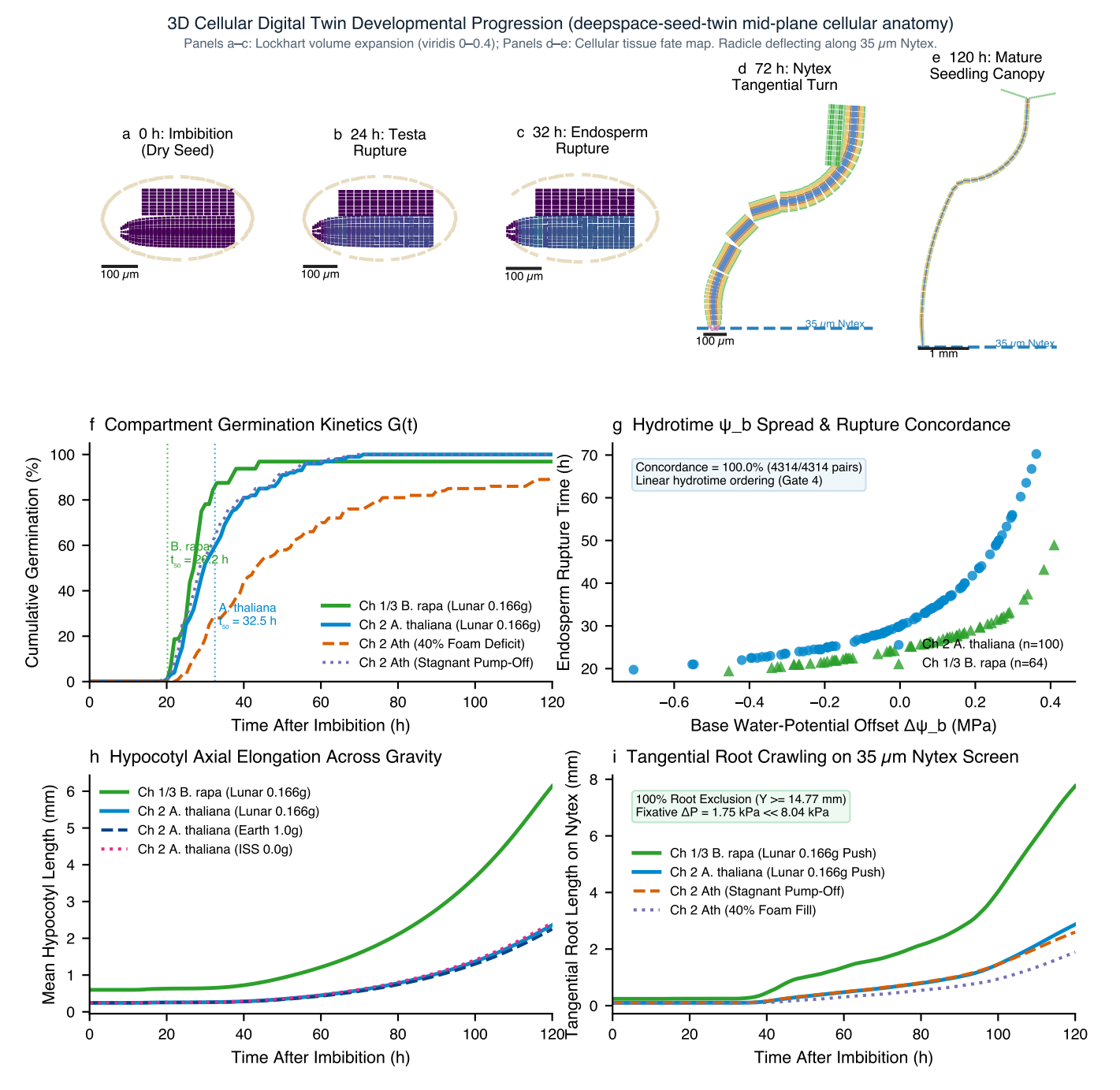

3D Cellular Digital Twin Developmental Progression (deepspace-seed-twin mid-plane cellular anatomy)
Panels a–c: Lockhart volume expansion (viridis 0–0.4); Panels d–e: Cellular tissue fate map. Radicle deflecting along 35 μm Nytex.
a 0 h: Imbibition (Dry Seed)
b 24 h: Testa Rupture
c 32 h: Endosperm Rupture
d 72 h: Nytex Tangential Turn
e 120 h: Mature Seedling Canopy
100 μm
35 μm Nytex
1 mm
f Compartment Germination Kinetics G(t)
Cumulative Germination (%)
Time After Imbibition (h)
B. rapa
A. thaliana
Ch 1/3 B. rapa (Lunar 0.166g)
Ch 2 A. thaliana (Lunar 0.166g)
Ch 2 Ath (40% Foam Deficit)
Ch 2 Ath (Stagnant Pump-Off)
g Hydrotime ψ_b Spread & Rupture Concordance
Concordance = 100.0% (4314/4314 pairs)
Linear hydrotime ordering (Gate 4)
Endosperm Rupture Time (h)
Base Water-Potential Offset Δψ_b (MPa)
Ch 2 A. thaliana (n=100)
Ch 1/3 B. rapa (n=64)
h Hypocotyl Axial Elongation Across Gravity
Mean Hypocotyl Length (mm)
Time After Imbibition (h)
Ch 1/3 B. rapa (Lunar 0.166g)
Ch 2 A. thaliana (Lunar 0.166g)
Ch 2 A. thaliana (Earth 1.0g)
Ch 2 A. thaliana (ISS 0.0g)
i Tangential Root Crawling on 35 μm Nytex Screen
100% Root Exclusion (Y >= 14.77 mm)
Fixative ΔP = 1.75 kPa << 8.04 kPa
Tangential Root Length on Nytex (mm)
Time After Imbibition (h)
Ch 1/3 B. rapa (Lunar 0.166g Push)
Ch 2 A. thaliana (Lunar 0.166g Push)
Ch 2 Ath (Stagnant Pump-Off)
Ch 2 Ath (40% Foam Fill)